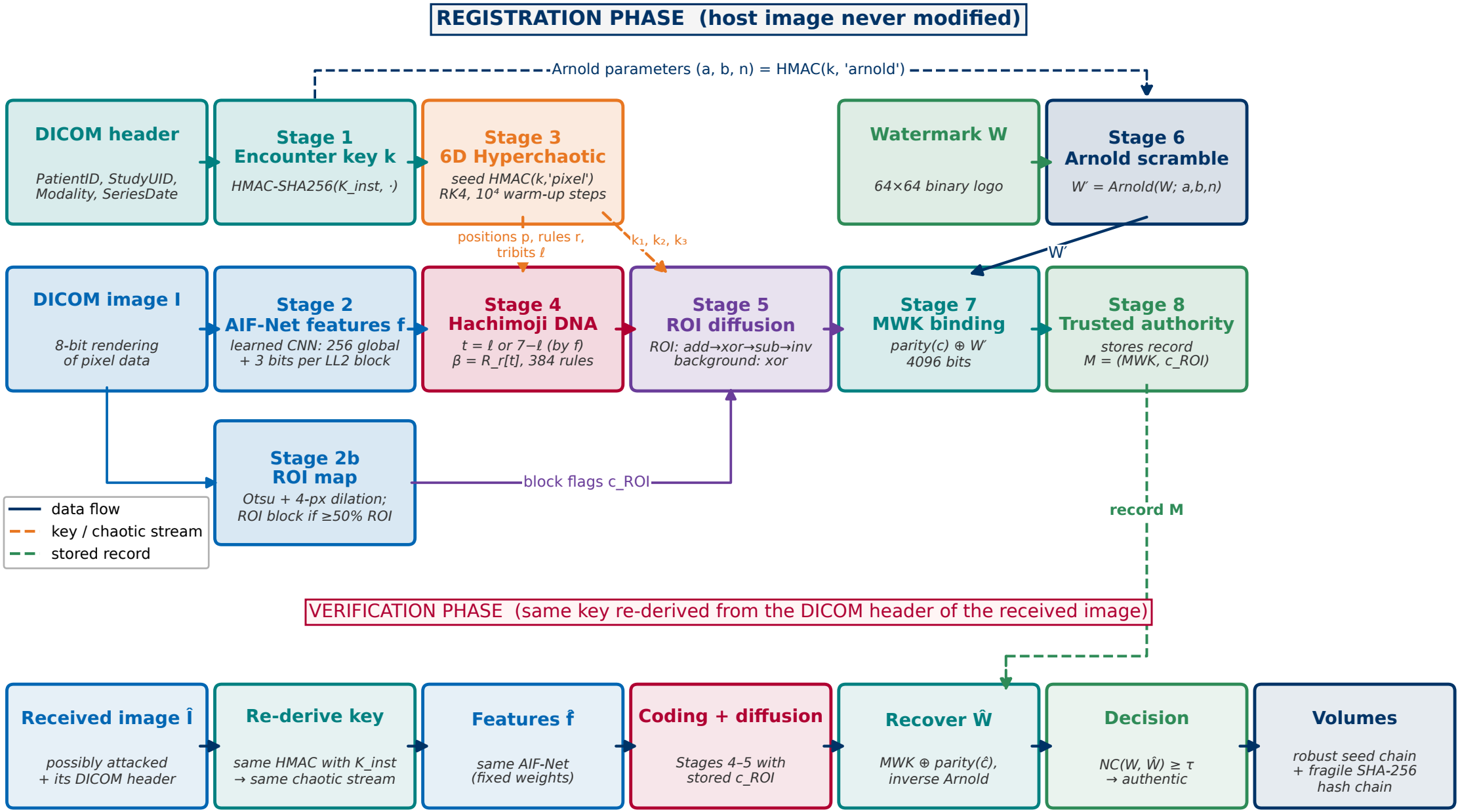

REGISTRATION PHASE (host image never modified)
Arnold parameters (a, b, n) = HMAC(k, 'arnold')
DICOM header
PatientID, StudyUID, Modality, SeriesDate
Stage 1
Encounter key k
HMAC-SHA256(K_inst, ·)
Stage 3
6D Hyperchaotic
seed HMAC(k,'pixel')
RK4, 10⁴ warm-up steps
Watermark W
64×64 binary logo
Stage 6
Arnold scramble
W′ = Arnold(W; a,b,n)
positions p, rules r, tribits ℓ
k₁, k₂, k₃
W′
DICOM image I
8-bit rendering of pixel data
Stage 2
AIF-Net features f
learned CNN: 256 global + 3 bits per LL2 block
Stage 4
Hachimoji DNA
t = ℓ or 7−ℓ (by f)
β = R_r[t], 384 rules
Stage 5
ROI diffusion
ROI: add→xor→sub→inv
background: xor
Stage 7
MWK binding
parity(c) ⊕ W′
4096 bits
Stage 8
Trusted authority
stores record
M = (MWK, c_ROI)
Stage 2b
ROI map
Otsu + 4-px dilation;
ROI block if ≥50% ROI
block flags c_ROI
record M
data flow
key / chaotic stream
stored record
VERIFICATION PHASE (same key re-derived from the DICOM header of the received image)
Received image Î
possibly attacked + its DICOM header
Re-derive key
same HMAC with K_inst → same chaotic stream
Features f̂
same AIF-Net (fixed weights)
Coding + diffusion
Stages 4–5 with stored c_ROI
Recover Ŵ
MWK ⊕ parity(ĉ), inverse Arnold
Decision
NC(W, Ŵ) ≥ τ → authentic
Volumes
robust seed chain + fragile SHA-256 hash chain